Individual Assignment 5

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Set Up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visulaize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data","taxon_list.csv"))
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
  )

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c(" ", "n/a", "N/A", "over", "173+")
)
```

Cleaning Data

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS quaerterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join to only include sites in ncos
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with data frames
```

```r
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

```r
aq_ins_clean <- aq_ins |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  rename (date = date_on_vial) |>
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  filter(sample_type %in% c("FB250", "FB 250")) |>
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
    ) |>
  group_by(date, site, taxon) |>
  summarize(ave_lit = sum(abundance, ra.rm = TRUE)/7.5) |>
  ungroup() |>
   # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
```

```r
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  ))
```

```r
# base layer
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # different theme
  theme_bw()
```

```r
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```r
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()
```

1. Visulaizing differences across sites in major taxon abundance

a)

- x-axis: site
- y-axis: log + 1 transformed average abundance/L
- geom to represent average abundance/L: geom_jitter()
- geom to represent medians: stat_summary()

b)

```r
# creating new object from clean aquatic inverts
ostracod <- aq_ins_clean |>
  # filter to only include ostracod
  filter(taxon == "ostracod") |>
```

```r
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

ostracod |>
  slice_sample(n = 10)
```

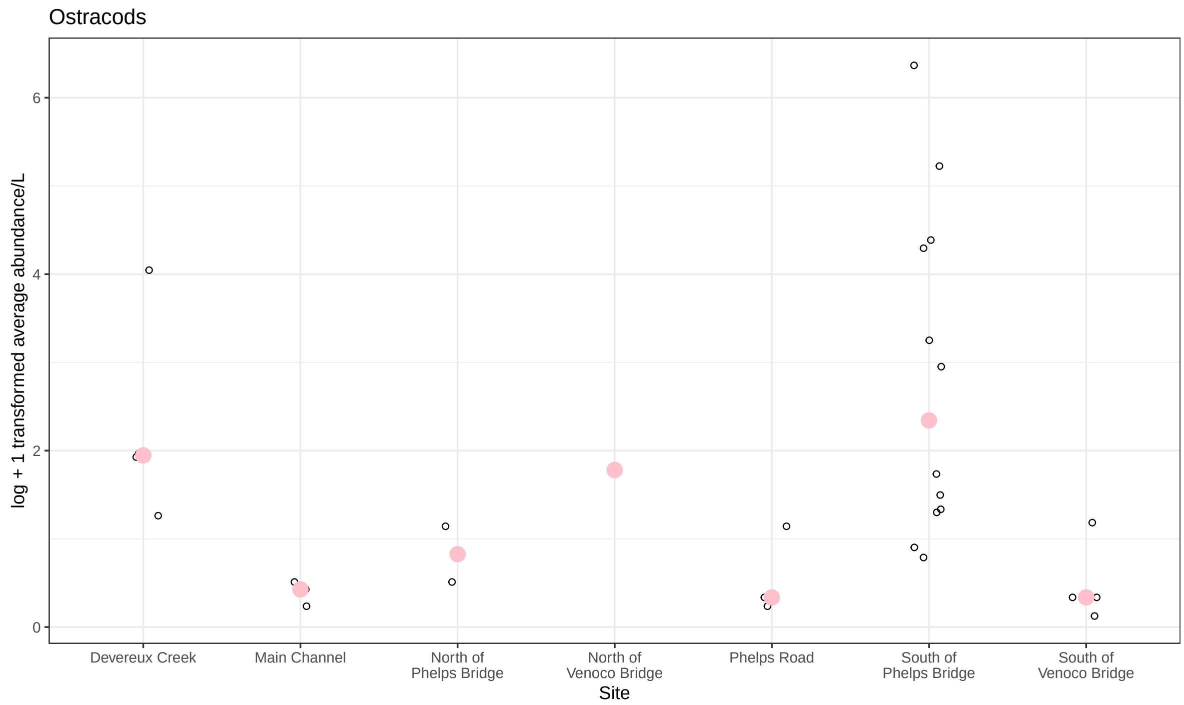
```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-03-03_NDC  2023-03-03 00:00:00 NDC   ostr~    5.87   7.45   0.688   5.64
 2 2022-08-12_NVBR 2022-08-12 00:00:00 NVBR  ostr~   NA     NA     NA      NA
 3 2022-11-12_NEC  2022-11-12 00:00:00 NEC   ostr~   NA      7.33   7.01    2.4
 4 2022-01-21_NEC  2022-01-21 00:00:00 NEC   ostr~   NA     NA     NA      NA
 5 2023-05-05_NPB  2023-05-05 00:00:00 NPB   ostr~   24.8   NA     NA      12.1
 6 2023-02-21_NEC  2023-02-21 00:00:00 NEC   ostr~   NA      8.56  NA       8.8
 7 2023-05-06_NDC  2023-05-06 00:00:00 NDC   ostr~   56.1   10.1   NA      13.1
 8 2023-09-12_NVBR 2023-09-12 00:00:00 NVBR  ostr~   NA      9.14  NA      20.3
 9 2021-11-23_NPB2 2021-11-23 00:00:00 NPB2  ostr~    0.4    7.59   1.99    0.65
10 2023-11-14_NPB  2023-11-14 00:00:00 NPB   ostr~   18.1   NA     NA      NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <chr>, log_ave_lit <dbl>
```

c)

```r
# base layer
ostracod_plot <- ggplot(data = ostracod,
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "pink") +
  # wrapping x-axis text
```

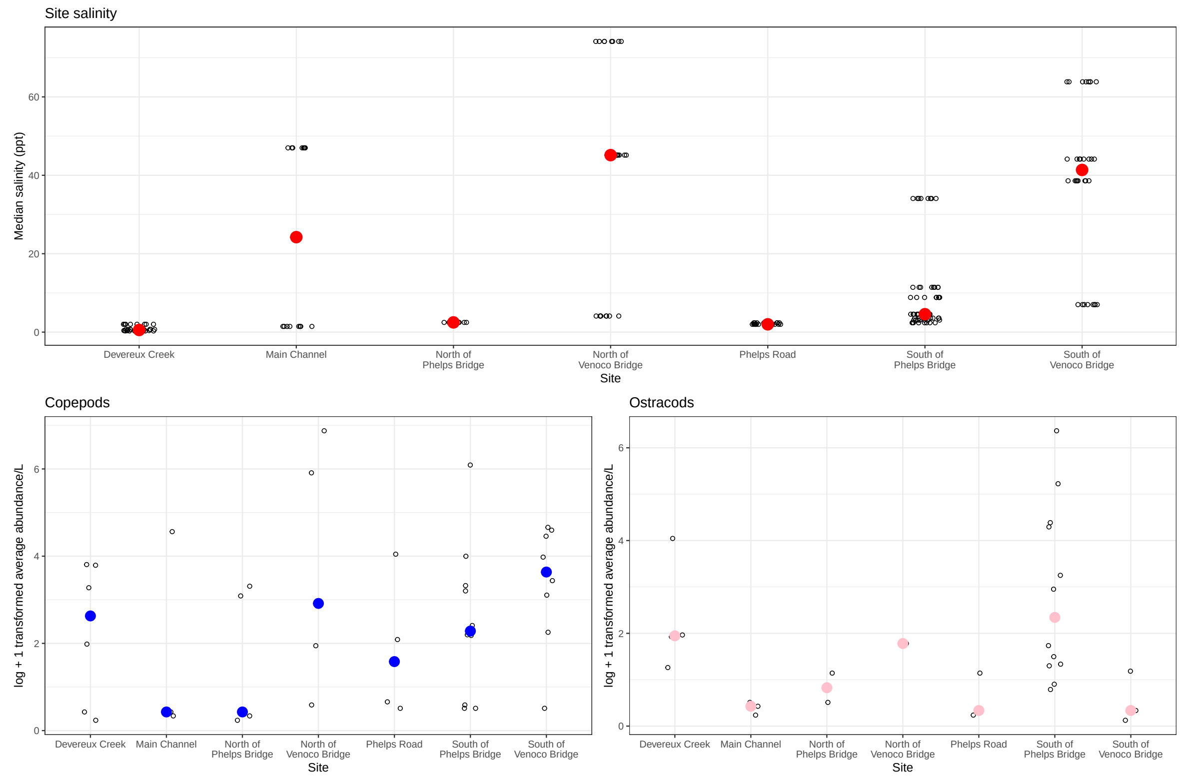
```r
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # different theme
  theme_bw()

# displaying object
ostracod_plot
```



d)

```r
# create multipanel figure
salinity / (copepods + ostracod_plot)
```



e)

Copepods are more abundant than ostracods at these sites: Devreux Creek, North of Venoco Bridge, Phelps Road, and South of Venoco Bridge.

It does not appear that ostracod abundance relates to site-level differences in salinity. At South of Venoco Bridge salinity is high while abundance is low however at North of Venoco Bridge both salinity and abundance are high.

2. Visualizing total abundance as a function of salinity

a)

- x-axis: med_sal
- y-axis: log_ave_lit
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~ taxon, scales = "free")

b)

```
# creating new object from clean aquatic inverts
aq_ins_salinity <- aq_ins_clean |>
  # create log + 1 transformed average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1),
         # make taxon names title case
         taxon = str_to_title(taxon)) |>
  # order taxon levels by maximum abundance
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep columns needed for figure
  select(taxon, ave_lit, log_ave_lit, med_sal, site_full)

# display 10 random rows
aq_ins_salinity |>
  slice_sample(n = 10)
```

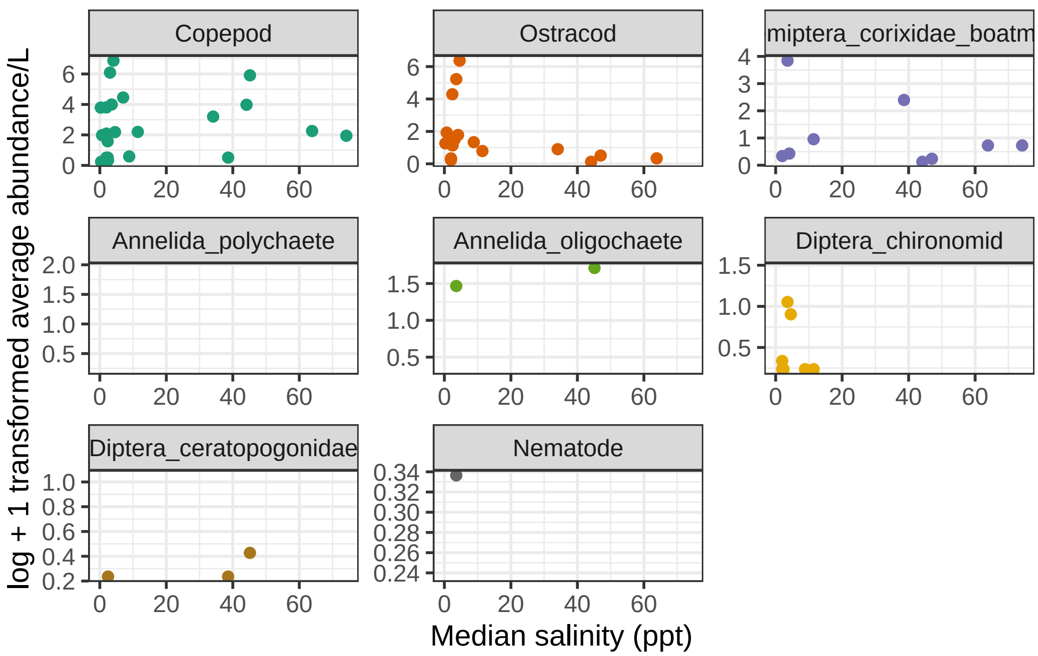
```
# A tibble: 10 x 5
   taxon                         ave_lit log_ave_lit med_sal site_full
   <fct>                           <dbl>       <dbl>   <dbl> <chr>
 1 Nematode                           NA       NA       2.47 North of Phelps Brid~
 2 Ostracod                           NA       NA      NA    North of Venoco Brid~
 3 Hemiptera_corixidae_boatman        NA       NA      NA    North of Phelps Brid~
 4 Nematode                           NA       NA      NA    Main Channel
 5 Diptera_chironomid                 NA       NA      NA    Main Channel
 6 Annelida_polychaete                NA       NA      34.1  South of Phelps Brid~
 7 Annelida_polychaete                 6        1.95   NA    South of Phelps Brid~
 8 Annelida_polychaete                NA       NA       0.3  Devereux Creek
 9 Copepod                            NA       NA      NA    North of Venoco Brid~
10 Annelida_oligochaete               NA       NA       0.398 Devereux Creek
```

c)

```
salinity_taxa <- ggplot(data = aq_ins_salinity,
                        mapping = aes(x = med_sal,
                                      y = log_ave_lit,
                                      color = taxon)) +
  geom_point() +
  # separate panel for each taxon
```

```
  facet_wrap(~ taxon, scales = "free") +
  # custom colors
  scale_color_brewer(palette = "Dark2") +
  # labels
  labs(x = "Median salinity (ppt)",
       y = "log + 1 transformed average abundance/L") +
  # remove legend
  theme_bw() +
  theme(legend.position = "none")

# display figure
salinity_taxa
```



d)

For Annelida Polychaete, Annelida Oligochaete, and Nematode, not much can be said for the relationship between abundance and salinity due to a lack of data. For Copepod and Ostracods it seems that both species are the most abundant at low mean salinities. It also seems that Copepods are more resiliant to higher salinities because their abundance remains high even past 60 ppt while Ostracod abundance decreases. Diptera Chironomid are only abundant at low salinities and Boatmen have similar abundances across all salinities.